

Sameer Shashikant Rajendra

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SUMMARY

ML/AI Engineer with 3 years building production computer-vision and LLM systems, and deep hands-on depth across the full inference stack: CUDA decode kernels, FP8 KV-cache quantization, PyTorch training, LangGraph agents, RAG, and self-hosted vLLM/SGLang serving. Ships models end to end — data pipelines, distributed training, quantization, serving, and evaluation — on AWS with Docker, Terraform, and CI. M.S. in Applied Artificial Intelligence from Stevens (Computer Engineering concentration, May 2026). Optimized a Hopper attention kernel 10.8× and served a 7B model at 4,699 tokens/sec on a single H100.

TECHNICAL SKILLS

Languages: Python, C++, CUDA C/C++, SQL, Bash

Machine Learning & Deep Learning: PyTorch, TensorFlow, JAX, scikit-learn, XGBoost, NumPy, pandas, Hugging Face Transformers, LoRA / fine-tuning, mixed precision, distributed training (FSDP, NCCL), model calibration

LLM & Generative AI: RAG, vector search (pgvector, HNSW), embeddings, LangGraph, agentic workflows, tool calling, Model Context Protocol (MCP), structured outputs, prompt engineering, LLM evaluation (LLM-as-judge, offline and adversarial evals), vLLM, SGLang, quantization (INT8 / INT4 / FP8)

MLOps & Infrastructure: Docker, AWS (Lambda, DynamoDB, S3, EventBridge, SNS, CloudWatch, ECR), Terraform / Infrastructure-as-Code, CI/CD, Git, Linux, SLURM, FastAPI, REST APIs, observability (Langfuse), serverless & event-driven architecture

GPU & High-Performance Computing: CUDA, cuDNN, CUTLASS, NCCL, roofline analysis, profiling (Nsight Systems / Compute)

Computer Vision: CNNs, EfficientNet, object detection, pose estimation, OpenCV, TorchVision, GPU-accelerated preprocessing

EDUCATION

Stevens Institute of Technology, Hoboken, NJ

May 2026

M.S. in Applied Artificial Intelligence — Computer Engineering Concentration

Relevant coursework: GPU & Multicore Programming (CUDA, software–hardware co-design), Digital & Computer Systems Architecture, Real-Time & Embedded Systems, Applied Modeling & Optimization, Pattern Recognition & Classification

Symbiosis Skills and Professional University, Pune, India

May 2021

B.Tech. in Mechatronics

Relevant coursework: Digital Signal Processing, Communication Systems, Control Engineering, Microprocessors & Applications, Digital Hardware Design & Analysis

PROFESSIONAL EXPERIENCE

Oneirix Labs Pvt. Ltd. — Pune, India

Sept 2021 – June 2024

Associate Engineer (Computer Vision & Embedded Systems)

- Built and deployed a **multi-task computer-vision system on live Pune Metro CCTV**, running **4 concurrent PyTorch inference tasks** — crowd density, pose-based track-jump intrusion, property damage, and unattended baggage — for a public-transit client's operations team
- Owned the **controller–scheduler communication subsystem** for a smart traffic-light deployment — wrote the **bidirectional protocol decode/forward layer in C++**, containerized the automation stack with Docker, and ran on-site field integration and debugging alongside the client's engineers
- Built a **coronary-artery graph-traversal pipeline** for stenosis screening with **CNN-based main-vessel identification**, BFS traversal of secondary and tertiary branches, and geometric diameter-reduction detection
- Translated ambiguous client requirements into shipped systems across **transit, municipal infrastructure, and healthcare** engagements — owning scoping, implementation, deployment, and post-launch support

PROJECTS

GPU Systems Researcher — LLM Inference Optimization | [GitHub](#) | [Report](#)

Apr 2026 – May 2026

Independent Research

- Optimized a **split-KV FlashDecoding-style GQA decode kernel** (CUDA C++, Hopper sm_90a, Llama-3.1-8B) achieving **9.7× speedup at 64K context** (1.35 → 0.14 ms), raising HBM bandwidth utilization from 5.8% to **56% of peak** by indexing the grid on KV heads to cut redundant KV traffic 4×
- Built an **FP8 (e4m3) KV cache** with per-page scales dequantized in-register, halving KV bandwidth and residency to **double the concurrent 64K-context sequences an 80 GB H100 holds (7 → 14)**; gated at **+0.24% perplexity with 100% argmax parity** on confident predictions
- Benchmarked **five attention modes on real Llama-3.1-8B weights** at 4K–64K context, showing **per-layer dense/sparse routing restored 100% argmax parity** where uniform sparse attention scored 0%; every run SHA-stamped behind **28 correctness gates**

Accord — Negotiation Intelligence Platform | [GitHub](#)

June 2026 – July 2026

- Built the **evaluation and safety harness** for a non-deterministic LLM agent — deterministic citation-grounding, LLM-as-judge with position-bias controls, multi-tenant isolation, PII scanning, and prompt-injection suites — measuring a **1.8% citation-fabrication rate with 0 cross-namespace leaks**
- Architected a **LangGraph agent running 5 analysis stages in parallel** over a negotiation thread, grounded by retrieval against a **1,030-dialogue precedent corpus**, exposed identically over a **FastAPI REST API and an MCP tool server** from one shared implementation, with grammar-constrained structured outputs and per-stage graceful degradation
- Designed and ran a **controlled RAG-vs-no-RAG ablation** — LLM-as-judge with position-bias controls, **Wilson 95% confidence intervals**, and deterministic citation-grounding — built into the agent graph behind a single flag so retrieval shipped only on measured evidence
- Served **self-hosted Qwen2.5-7B on SGLang at 4,699 output tokens/sec** on a single H100 — **30× over single-stream** via continuous batching — at **\$0.27 per 1M output tokens** and **25 ms p50 time-to-first-token**, deployed scale-to-zero on Modal (~\$0 idle)

DeltaFin — Agentic Financial Variance & AP Reconciliation System | [GitHub](#) | [Live Demo](#)

Sept 2026

AI × Finance “Money Talks” Hackathon — Money Operations track

- Built a **10-node LangGraph variance-explanation agent** (plus a 7-node AP-reconciliation pipeline) that turns monthly account summaries and transaction subledgers into a **prioritized, owned variance brief**, with a hard deterministic/LLM boundary — Python computes every delta, z-score and materiality call, the model only writes prose over those finished facts
- Scored the pipeline against a **labeled 20-month ledger** with hand-transcribed ground truth, replaying **all 19 period comparisons** deterministically — **62 of 63 assertions pass**, and **36.0% detection precision** is published with the full unexplained-findings list rather than gated behind an unmeasured threshold

Adaptive Evolutionary GPU Optimization Framework (AEGOF) | [GitHub](#) | [Report](#)

Sept 2025 – Dec 2025

- Built a **multi-objective GPU autotuning framework** (NSGA-II via DEAP, $\mu+\lambda$ strategy) that jointly optimizes **8 hardware/software knobs** — power cap, batch size, sequence length, bf16 vs. fp16 precision, kernel fusion, block size, tile config, and FlashAttention-2 vs. CUTLASS backends — for Llama-2-7B inference on an NVIDIA H100
- Evolved a **throughput-vs-power Pareto front** across 16 individuals × 5 generations over **238+ configurations**, reaching an **8.09× efficiency multiplier** (tokens/sec/Watt) over PyTorch defaults, serialized to a Hall-of-Fame JSON as a reproducible release-qualification artifact

Serverless LLM-Scoring Pipeline (event-driven AWS) | [GitHub](#)

April 2026

- Built a **fully serverless, event-driven ingestion-and-scoring pipeline on AWS** — EventBridge cron triggers a containerized Lambda that scrapes sources hourly, dedupes idempotently against DynamoDB, applies a regex prefilter, then a **Claude scoring agent** classifies each survivor with SNS email alerting; the entire stack is defined in **Terraform** (Lambda, DynamoDB, S3, EventBridge, SNS, IAM, budgets) and deploys from a single terraform apply

AWARDS & RECOGNITION

HACK MIT-WPU 2024 — CCTV Video Analytics Hackathon

May 2024

- Maha Metro × Ministry of Education, Maharashtra. Built a **real-time multi-task CCTV analytics system** for Pune Metro, recognized for innovation in sustainable urban transit infrastructure